

Anatoly Zavyalov

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Research Interests

Differential privacy, graph algorithms, sublinear algorithms, property testing

Education

Boston University

Ph.D. Student in Computer Science — Advised by Sofya Raskhodnikova

Sep 2024 - Jun 2029 (expected)

University of Toronto

H.B.Sc. in Applied Mathematics — 3.95 cGPA

Sep 2020 - Jun 2024

Selected Publications

1. S. Raskhodnikova, A. D. Smith, C. Wagaman, **A. Zavyalov**, “Local Node Differential Privacy”. *In preparation*.
2. J. Shallit, **A. Zavyalov**. “Transduction of Automatic Sequences and Applications”. *CIAA 2023. Lecture Notes in Computer Science*, vol 14151.
3. A. D. Hincks, **A. Zavyalov**, D. Bansal. “A graph database solution for tracking the deployment and layout of a large radio interferometer”. *SPIE Astronomical Telescopes + Instrumentation 2022. Proc. SPIE* 12189.

Honors and Awards

Chair’s Graduate Fellowship - \$10,000 USD

Awarded on admission to Boston University.

Feb 2025

NSERC Undergraduate Student Research Award - \$7,500 CAD

Awarded by U of T Department of Computer Science.

Mar 2023

SURP Fellowship & U of T Excellence Award - \$17,095 CAD

Combined funding for a research fellowship at the Dunlap Institute of Astronomy.

Apr-May 2021

NSERC USRA (\$7,500 CAD) & Fields USRP — Declined

Received and declined offers for two separate summer research programs.

Mar 2022

University of Toronto Scholar - \$3,000 CAD

Awarded by the University of Toronto for high academic achievement (\$1,500 × 2).

Aug 2021, Aug 2022

Dean’s List Scholar

Awarded for a cumulative GPA of 3.50 or higher.

2021, 2022, 2023

Trinity College Academic Achievement Scholarships - \$1,500 CAD

Awarded by Trinity College for high academic achievement (\$500 × 3).

Nov 2021 - Nov 2023

Talks

1. “Transduction of Automatic Sequences and Applications”, CIAA 2023. (30 min) (Slides)
2. “Automatic Sequences”, Canadian Undergraduate Math Conference, 2023 (30 min) (Slides)
3. “Automatic Sequences”, UTSC CMS Undergraduate Seminar, 2023 (60 min) (Slides) (Recording)

Research Experience

Boston University | PhD Student

Sep 2024 - Present

Department of Computer Science

- Working in the areas of graph differential privacy and graph property testing, supported by a **Chair's Graduate Fellowship**.

University of Waterloo | Research Assistant

May 2022 – Jul 2022, May 2023 – Jul 2023

David R. Cheriton School of Computer Science

- Researched and implemented algorithms into **Walnut**, a theorem proving software for automatic sequences written in **Java**, working with Professor Jeffrey Shallit.
- Research culminated in a publication and presentation at *CIAA 2023*.

University of Toronto | Research Assistant

May 2023 – Aug 2023

Department of Computer Science

- Researched algebraic methods for concurrent program verification and race condition detection, supported by an **NSERC Undergraduate Student Research Award (NSERC USRA)**.

University of Toronto | Research Fellow

May 2021 – Apr 2022

David A. Dunlap Department of Astronomy and Astrophysics

- Developed **Padloper**, a full-stack graph database solution for tracking deployment and layout of a large radio interferometer, using **JanusGraph**, **Flask** and **React**, working with Professor Adam Hincks.
- Padloper will be used for the **Hydrogen Intensity and Real-time Analysis eXperiment (HIRAX)**, the **Simons Observatory**, and **Canadian Hydrogen Intensity Mapping Experiment (CHIME)**.
- Research culminated in proceedings at *SPIE Astronomical Telescopes + Instrumentation 2022*.

Teaching Experience

Boston University | Teaching Fellow

- Teaching Fellow for **CSC237: Probability in Computing**

Sep 2025 - Present

University of Toronto | Teaching Assistant

- TA for **CSC363: Computational Complexity and Computability**

Jan 2024 - Apr 2024

- **Lead** TA for **CSC373: Algorithm Design and Analysis**

Sep 2023 - Dec 2023

- TA for **CSC240: Enriched Introduction to the Theory of Computation**

Feb 2022 - May 2022

STEM Outreach

SigmaCamp | Academic Administrator & Counselor

2022 - Present

- Directing *Problem of the Month*, a free international STEM competition for teenagers, which received over 2,000 solution submissions from 130+ students in 2024-2025.
- Assembled and managed a team of 12 subject leads and dozens of graders. Also managed the Qualification Quiz entrance exam, which received 650+ solutions submissions in 2025.

Skills

- **Human Languages:** English (C2), Russian (C2), German (B1), Portuguese (A2)
- **Programming Languages:** Python, JavaScript, C++, Java, PostgreSQL, $\text{\LaTeX}$
- **Frameworks/Libraries:** ReactJS, Flask, NumPy, PixiJS